

REST API Reference Manual

REST API Reference Manual: Automated Reconciliation Engine

1. Overview

The Reconciliation API is built using **FastAPI** and runs on port `8000`. All requests must supply the appropriate headers for authentication if enabled.

2. Endpoints Summary

2.1 File Ingestion

```
* POST /api/v1/files/upload`
* Ingests bank statement or internal ledger files. Calculates file SHA-256 hash to prevent duplicate uploads.
* Form Data:
* `bank_code`: Bank config code (e.g. `HDFC`, `ICICI`, `AXIS`)
* `format_type`: File format (`CSV`, `MT940`, `CAMT053`)
* `source_type`: Source indicator (`INTERNAL` or `EXTERNAL`)
* `file`: Ingestion payload (multipart file)
* Response (200 OK):
{
  "status": "SUCCESS",
  "file_hash": "2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824",
  "filename": "hdfc_statement.csv",
  "records_ingested": 125,
  "anomalies_detected": 2,
  "parse_errors_count": 0,
  "errors": []
}
```

2.2 Execute Matching Pipeline

```
* POST /api/v1/reconciliation/run`
* Triggers the sequential matching engine (Exact -> Fuzzy -> Rule-Based) for a bank and run date.
* Request Body (JSON):
{
  "bank_code": "HDFC",
  "run_date": "2026-07-01"
}
* Response (200 OK):
{
  "id": "e81d77a0-0bbf-4630-9b88-1d2be11802a4",
  "bank_code": "HDFC",
  "run_date": "2026-07-01",
  "status": "COMPLETED",
  "total_records": 250,
  "matched_records": 210,
  "exception_records": 40,
  "match_rate": 84.00,
  "processing_time_ms": 125,
}
```

```
"created_at": "2026-07-01T12:00:00.000Z"
}
```

2.3 Exception Management

```
* **GET** `/api/v1/exceptions`
* Queries exceptions queue.
* **Query Parameters**:
* `status`: `OPEN`, `UNDER_REVIEW`, `RESOLVED`
* `severity`: `LOW`, `MEDIUM`, `HIGH`, `CRITICAL`
* `category`: `MISSING_INTERNAL`, `DIRECTION_REVERSAL`, etc.
* `bank_code`: `HDFC`, etc.
* **Response (200 OK)**:
[
  {
    "id": "90e66d40-cc5c-42cb-b1b1-bd2cc584b42b",
    "run_id": "e81d77a0-0bbf-4630-9b88-1d2be11802a4",
    "normalised_txn_id": "7fa12bf1-99bc-4ad2-bf01-cb993d98be22",
    "category": "DIRECTION_REVERSAL",
    "severity": "CRITICAL",
    "status": "OPEN",
    "assigned_tier": 4,
    "sla_deadline": "2026-07-01T12:30:00.000Z",
    "created_at": "2026-07-01T12:00:00.000Z"
  }
]

* **PUT** `/api/v1/exceptions/{id}/resolve`
* Resolves an open exception.
* **Request Body (JSON)**:
{
  "resolution_type": "MANUAL_MATCH",
  "resolution_details": {
    "matching_txn_id": "89ab3cdf-77bc-4ad9-bf90-cb883d98be11",
    "reason": "Operator matched UTR manually after bank verification"
  }
}
```

2.4 Cryptographic Audit Trail

```
* **GET** `/api/v1/audit`
* Fetches list of all system audit logs.
* **GET** `/api/v1/audit/verify`
* Re-computes and checks all SHA-256 chain links to verify integrity.
* **Response (200 OK)**:
{
  "status": "SECURE",
  "integrity_verified": true,
  "tampered_records_count": 0,
  "violations": []
}
```

2.5 Real-time Dashboard Summary

```
* **GET** `/api/v1/dashboard/summary`
* Fetch dashboard KPIs, categorisation charts, and rolling 7-day trend.
```